Yousuf Shahabuddin

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I'm a 3rd year Design Engineering MEng student at Imperial College London, graduating June 2028. Most of my work so far comes down to testing whether something can be trusted: validating simulation models against independent references, and building and testing hardware for a competition vehicle. Available late March to September 2027, with J-1 visa sponsorship arranged through CCUSA.

Education

Design Engineering, MEng, Imperial College London 2024 – 2028
- 3rd Year Student. Modules: Product Design Engineering, FEA, Thermofluids, Human-Centred Design, Behavioural Changes, Sustainable Design Engineering, Electronics, Solid Mechanics, Materials & Manufacturing, Computing, Data Science, Entrepreneurship, Nano Design.
- Predicted 2:1 (Upper Second with Honours)

Secondary School, King Edward VI Grammar School, Chelmsford 2017 – 2024
- A-Levels: Maths, Physics, Design & Technology, Chemistry, EPQ

EPQ: “To what extent can Supersonic Commercial Aviation be Revived?”

Experience

Co-founder, FemAI 10/2025 – Present
- A B2B company aimed at improving IVF implantation success rates. Accelerator phase.
- Researched IVF clinics and regulations, and was responsible for pitch deck design.
- Awarded 1st place in the Women’s Longevity hackathon. Currently in progress of launching the company and partnering with IVF clinics.

Automotive Mechanic, Imperial College Eco Marathon Team 10/2025 – Present
- Refining the pedal and throttle system using anthropometric data to optimise driver comfort and control.
- Working on tyre fixtures and improving chassis assembly to reduce disassembly time.

Freelance Tech Procurement 08/2024 – Present
- Developed my own frameworks for tech purchasing based on budget, performance, and user specific needs.
- Successfully advised numerous students, family members, and other individuals on purchasing technology that best suits their specific needs and budgets, specialising in computers, phones and monitors.

Mathematics, Science, and Design Tutor 12/2022 – Present
- Honed my ability to explain complex concepts in a clear and concise manner to students aged 12-18.
- Built strong rapport and interpersonal skills through one-on-one tutoring sessions, fostering a positive and encouraging learning environment.

EPSRC Intern 06/2026 – 08/2026

Lumped Parameter Modelling of Li-ion Batteries
- Investigated parameter identifiability of lumped-parameter lithium-ion battery models; built pypline, a Python toolchain for local sensitivity analysis, Fisher information, parameter scaling and grouping, and automated reporting and documentation.
- Raised the number of identifiable parameters from 1 to 2 of 11 to 4 to 8 of 10 across a state-of-charge by C-rate experimental grid by re-parameterising into identifiable groups.
- Validated the model against an independent open-source reference implementation to under 1 mV agreement; identified and corrected three parameter transcription errors in the research group’s simulation package.
- Designed and deployed a Sobol global sensitivity analysis pipeline for HPC execution, applying statistical variance decomposition to rank parameter influence across the full model.
- Python, NumPy, SciPy, PyBaMM, Git, HPC batch scheduling.

Tech Oversight, Team Assistant, FirstGenDoctors 07/2025 – 09/2025
- Designed & built their new website to increase engagement with their services and community.
- Collaborating with multidisciplinary teams to launch scalable outreach initiatives, aiming to help students and professionals advance within and enter the medical world.
- Leading strategy, idea generation, and event planning.

Summer Engineering Experience, InvestIN 08/2023 – 08/2023
- Designed & tested aerofoils to achieve a maximum angle of attack using a water tunnel.
- Coordinated work within my team ensuring our idea was well thought out and rehearsed ahead of the presentation.
- Won 1st place for my pitch on EV recharging and energy redistribution concept within developing cities.

Work Experience, LEGO Group 08/2022 – 08/2022
- Conducted user research for mobile UX, for future LEGO website designs. Experimented on Miro.
- Created and implemented processes, boosting LEGO’s social media engagement.
- Contributed to circular economy meetings, suggesting sustainable initiatives to implement within LEGO.

- Published meta descriptions for major LEGO products and categories to increase engagement with products.

Projects

Portfolio Website Creation, design.yousufshahabuddin.com
- Designed and built my own portfolio website. Hosted using GitHub pages. Project in progress as I experiment with designs (layout, aesthetics, and content).
- Currently testing future layouts on Figma and converting them to code to implement.

British Airways Data Science Job Simulation, Forage
- Completed a virtual project simulating the work of a BA Data Science team.
- Analysed customer review data to identify insights and trends.
- Built a predictive model in Python to understand buying behaviour.
- Developed PowerPoint deliverables to communicate findings to a business audience.

Chimp Game & "Martian" Hexadecimal Communication Robot
- Recreated a robotic system inspired by "The Martian" to point to hex characters; simulated Earth–Mars relay.
- Created a Chimp memory game with LED sequences accessed via an app.

NEXUS — Multiplayer Physical Dance Game for Concert/Event Queues
- Designed a large-screen motion-tracking dance game to foster engagement between people in concert queues.
- Virtual dance characters modelled, game programmed with MediaPipe and Unity.
- Extensive research and prototyping carried from idea to final prototype.
- Contributed to making our presentation slides and pitched our product.

R.C. Glider & Collision-Avoidance Robot
- Installed and tested sensors and controls to operate the RC glider.
- Created and coded a crash-avoidance robot intended for use in small crawl spaces.

Drinks Dispenser for Café ("MEDD", Jeddah)
- Created a drinks dispenser for my client "MEDD" café, a small chain in Jeddah, Saudi Arabia, to allow baristas to focus on serving specialty coffee, rather than serving customers who ask for pre-made beverages.
- Modular drip tray for "hot-swappable" aesthetics, cleaning and repairs. Magnetic opening to allow for easy access to staff for maintenance.
- Woodworking to produce the final product. Sketched, modelled with CAD, prototyped.

Leadership Experience, Skills, Interests

University Tours
I deliver tours to offer holders and prospective students; represented my department and ensured all questions from prospective students and their families were answered and well informed about the university.

Scouts Leader
- Assumed a key role in developing and implementing the curriculum, ensuring the smooth running of weekly sessions. Designed weekly curriculum, mentored Scouts, and ran sessions.
- Awarded Jack Petchey Award for outstanding contribution.

Skills
Project Management, Figma, Fusion360, Ansys Structural & Fluent, Python, MATLAB, Javascript, C++.

Interests
NICAS certified lead climber, Archery, Free Diving, Duke of Edinburgh Gold, Global Travel Planner, Airline commercial operations, Aircraft Design, Karting.